**Czech 20-month olds interpret gender-marked adjectives without nouns, showing emerging knowledge of the gender agreement system**

Children show early knowledge of grammatical gender, e. g. by using determiner gender to anticipate the upcoming nouns[1]. This could be due to learning the lexically specific relation between a noun and a determiner. Yet children need to form an abstract category of gender and gender agreement to acquire productive gender marking on agreeing elements. In languages that show few word-form cues about noun gender (e. g. Slavic languages), the agreement also serves as the main source of information for learning individual noun gender. Comprehension of productive gender morphology has been shown in Italian 12- and 18-month-olds, using gender variants of nouns, such as ragazzo/ragazza[2]. This could be interpreted as lexical knowledge connecting a form and a referent. In our experiment, we examined the knowledge of the gender system by presenting young children with gender-marked adjectives rather than nouns, avoiding direct reference of the marked element to a picture and relying on the gender inflection only.

Children were shown pairs of face photos (male/female), and their gaze direction was followed with the Eyelink 1000+ eyetracker. Children heard short phrases describing the pictures, with two instances of a gender-marked adjective (one of *smart, happy, sad* and *pretty*):

Jéé, hele! Koukej, jak je chytrý/chytrá! Je fakt chytrý/chytrá!
Yay, see! Look how smart$_{masc/fem}$ (s/he) is? (S/he) is really smart$_{masc/fem}$!

A total of 16 items were presented, 8 in each gender condition, using 4 distinct pairs of faces. Each pair was shown two times in each gender condition, counterbalancing the placement of the male face and order of presenting the masculine/feminine condition. Two baseline, asking to look at “pán/paní” (man/lady), showed clear discrimination between faces.

After excluding trials with more than 25% of data loss, we analyzed data from 32 children aged 18 to 24 months (M=20.52). We first evaluated the proportion of looks towards the male faces between 1 and 3 seconds after each adjective onset (see Fig 1). Then we used permutation analyses of binomial mixed models evaluating looks to male faces in 20-ms intervals from 1 s before to 3 s after the adjective onset. There was no significant difference between conditions in the proportion after the first adjective ($t=0.45$, $p=0.651$), but the effect was significant after the second adjective ($t=2.16$, $p=0.032$). Also the permutation analysis found no significant sequence of differences after the first adjective but a significant cluster from 1760 to 2280 ms after the second adjective onset (sumary $t=61.87$, $p=0.033$).

The results indicate that Czech children around 20 months of age show emerging sensitivity to gender inflections on adjectives, at an age when they typically produce no adjectives or adjectival agreement. They thus have some knowledge of grammatical gender as a category that is not tied to individual nouns. This also shows knowledge of morphological gender marking in adjectives, and of the connection between grammatical and natural gender. The effect only appears slowly after repeated gender marking, suggesting that the knowledge and processing system are early in in the process of their formation.

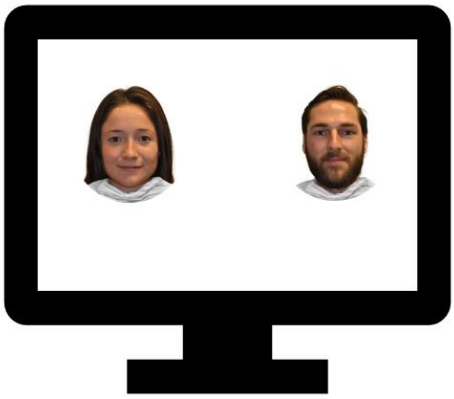

Figure 1: The scheme of a trial setup.

Jéé, hele! Koukej, jak je chytrý/chytrá! Je fakt chytrý/chytrá!
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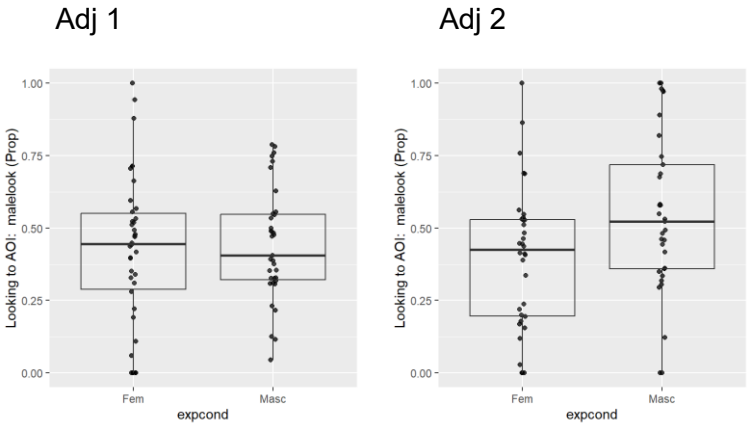
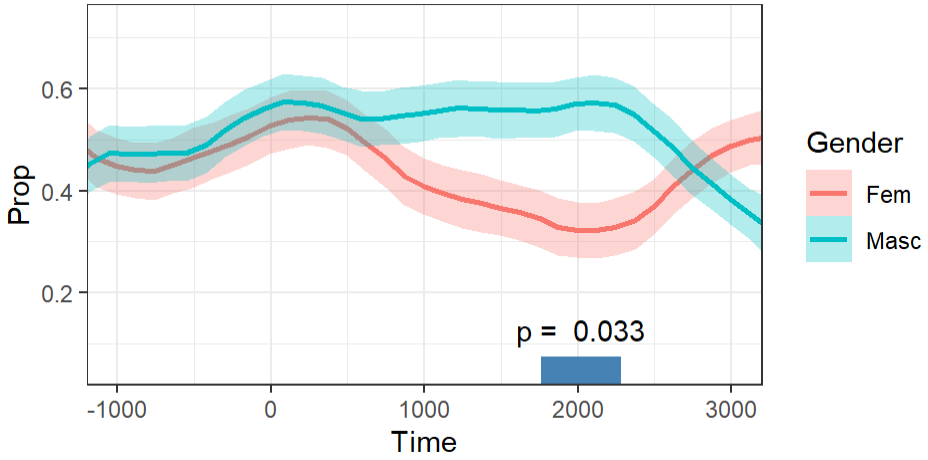



Figure 2: Proportions of looks to the male face per condition after the first and second adjective.

Figure 3: Time course of the proportion of looks towards the male face, from 1 s before to 3 s after the second adjective onset, with the significant cluster of differences marked.




## References


[1] Van Heugten & Shi, (2009). *Dev. Sci.* 12, 419–425. Smolík & Bláhová, (2019). *J. Exp. Child Psy.* 178, 251–265.
[2] Ferry, A., Nespor, M., Mehler, J. (2020), *Dev. Psy.* 56, 40-52.